A Analysis of Task Circuits

A.1 IOI Circuit & Algorithmic Criteria

To determine algorithmic consistency for the IOI circuit, we apply path patching as described in Appendix B in addition to using the component scores described in Appendix D. These are used to set thresholds for classifying attention heads. Though component score thresholds can be arbitrary, applying them consistently across all model checkpoints allows us to see the degree of similarity involved with model behavior.

Concretely, we use the following metrics and thresholds:

Direct-effect heads We initially perform path-patching on all model attention heads, measuring their impact on the logit different after the final layer of the model. We then classify attention heads as name-mover heads (NMHs), negative name-mover heads, and copy suppression heads (CSHs) based on copy score (for NMHs) or CPSA (for CSHs) of $> 10\%$, which yielded a small set of heads responsible for most of the direct effect. We measure the ratio of the absolute direct effect on logit difference for these heads vs. the total direct effect of all heads (including several unclassified heads) to obtain our first value.

Next, we conduct path-patching with NMHs as the receivers. This yields a set of heads that we then test for S2-inhibition (S2I) behavior, using Wang et al.'s [71] test for the effect of token signal vs. positional signal: does the ablation of these positional signal heads A). reduce the logit difference through the NMHs, B). reduce NMH attention (which determines what they copy) to the indirect object token, and C). increase attention to the subject tokens? If a head meets all of these conditions, we classify it as an S2I head, as it emits a signal used by the NMHs to decide what to copy. The total absolute effect of these heads on the NMHs is then divided by the total absolute effect of all heads on the NMHs, producing our second measurement.

Finally, we conduct path-patching with S2I heads as receivers. Here, we apply a simpler test since these heads can be quite diffuse throughout the model: Do the heads involved have above-average induction or duplicate-token scores? If so, we classify them as induction heads or duplicate token heads (confirming via manual examination of attention patterns and behavior), and divide the total absolute effect of these heads by the total absolute effect of all heads on the S2I heads, producing our third measurement.

These three metrics capture the extent to which known and classifiable model components contribute at each of the three primary levels of the IOI circuit. If the degree to which unknown or unclassified components contribute to any part of the circuit, we will see the corresponding score drop. As we see that in practice they tend to stay level, we conclude that there is a high degree of stability for this circuit.

B Other Circuit-Analysis Methods

Circuit analysis can be conducted via a number of different methods; the method used to find the original IOI circuit (and that we use to verify algorithmic consistency in this task) is Wang et al.'s [71] **path-patching**. Path patching is a specialized form of activation patching, used to isolate and analyze the influence of individual model components on a given task. Starting with two datasets (identical except for the key detail we want to base our circuit on, such as the correct and incorrect names in the IOI task), x_{orig} and x_{altered}, where x_{altered} is a counterfactual version of x_{orig}, the technique involves a sender attention head h and a set of receiver nodes $R \subseteq M$ within the model's computational graph M. Initially, activations are recorded from both datasets. Subsequently, all attention heads except h are locked to their activations from x_{orig}, while h is updated with its activation from x_{altered}. This configuration allows for a forward pass on x_{orig}, capturing intermediate activations for nodes $r \in R$. A final forward pass on x_{orig} then patches R to these stored values, facilitating the assessment of h's impact on the model's output.

Path patching aims to gauge the significance of the path $h \rightarrow r$ by comparing the model's logit differences across multiple pairs $(x_{\text{orig}}, x_{\text{new}})$. By averaging these differences over many pairs, the method effectively measures the impact of specific paths on model performance, providing insights into the contributions of individual components to the overall task. The process is iterative, such that

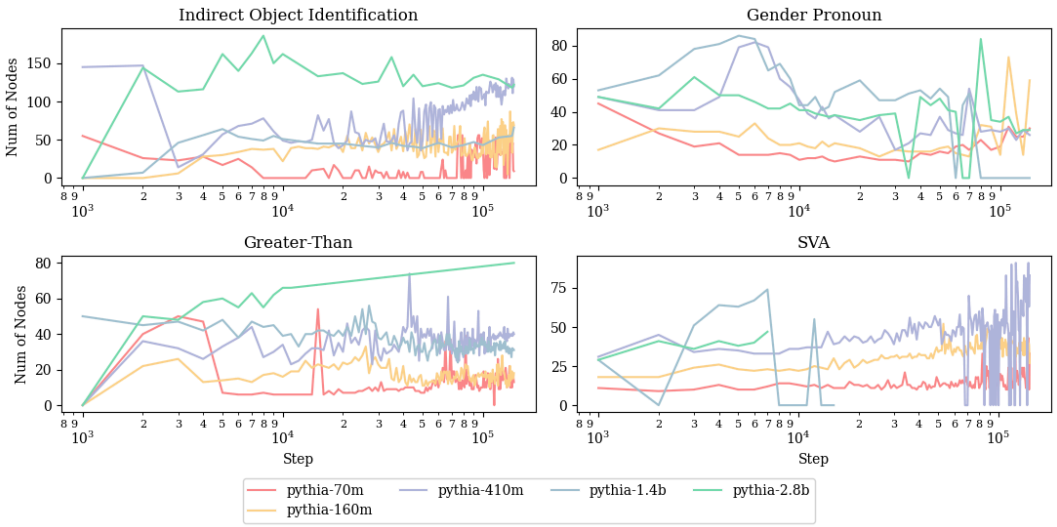

Figure 6: Number of Nodes in the circuits

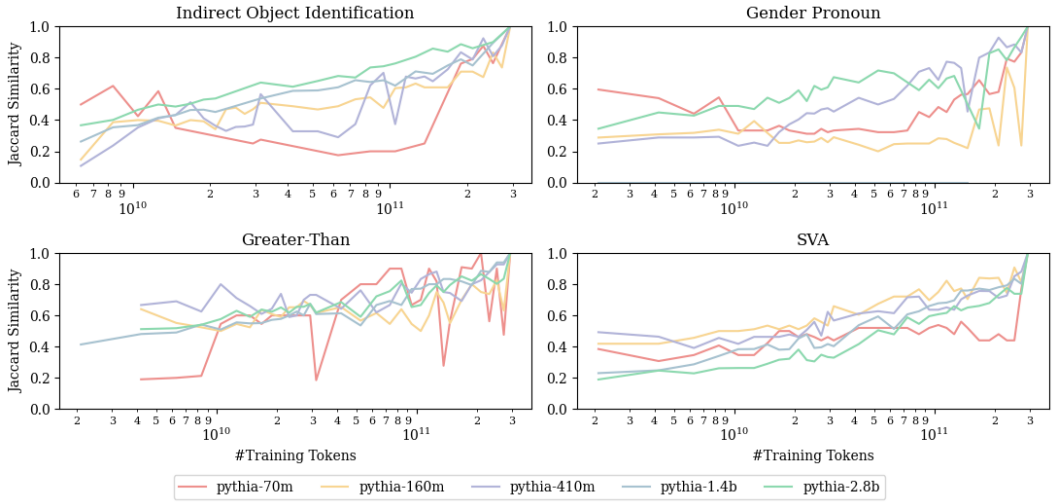

Figure 7: Node Jaccard Similarity of intermediate circuits with the circuit at the final checkpoint. Compared with the exponential weighted moving average (Figure 5, main text), we see more fluctuations, indicating that components swap during training at every checkpoint. Meanwhile, the upward trend indicates that circuits grow more and more similar to the final circuit during training.

a practitioner would start by observing which nodes impact the logits directly, and then proceeding backwards to see what nodes affect those first direct-effect nodes, and so on.

C Additional Size, Similarity & Change Rate Results

We graph out the number of nodes needed to generate faithful circuits across different checkpoints on all of the four tasks. Here we can observe that the number of nodes is positively correlated to the sizes of the models. When the model size increases the model needs more heads of the same kind to complete the same tasks. In the case of IOI, we can see the pink line for pythia-70m is at the bottom with the least number of nodes and the green line of 2.8b is at the top with the most number of nodes. This signifies a diffusion of the roles in attention heads. Increases in model size do not necessarily help heads become more specialized in their roles; rather, in these circuits more heads will take on the same roles.

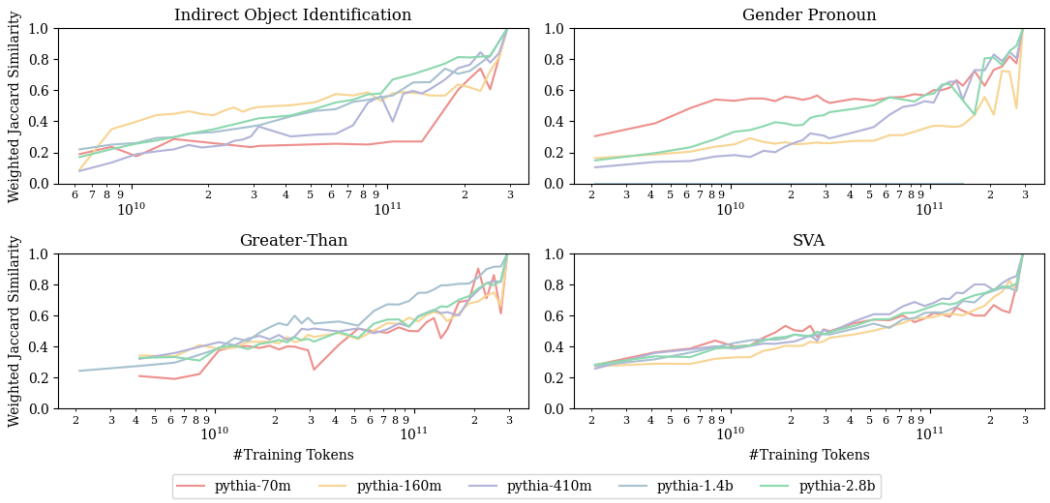

Figure 8: Edge Weighted Jaccard Similarity of the intermediate circuits with the circuit at the final checkpoint. The similarity slowly climbs and remains high at the end. Compared with the exact nodes, the weighted Jaccard Similarity of edges yields a smoother graph. It aligns with the conclusion that the model slowly drifts towards the final circuit along with components swapping during training

D Component Metrics

In this paper, we follow the metrics from previous literature in Wang et al. [71] for name-mover heads, McDougall et al. [45] for copy suppression heads, [54] for induction heads, and [29] for successor heads.

Copy Score Following Wang et al. [71], we check if the Name Mover Heads copy over the names across training time by using the same metrics- **copy score**. To validate the Name Mover Heads, we studied what values are written via the head's OV matrix. We take the state of the residual stream after the first layer of MLP on the specific name tokens. Then we multiply it with the OV matrix of the given heads, multiplied with the unembedding matrix and also the final layer norm. This simulates what will happen if the head attended perfectly to that token. We define copy score as the proportion of samples that contain the input name token in the top 5 logits.

CSPA Score McDougall et al. [45] introduced a novel approach named copy suppression-preserving ablation (CSPA), designed to ablate all behaviors of a specified attention head except for those related to copy suppression. This method involves two distinct types of ablation: OV ablation and QK ablation. In the OV ablation process, the output of an attention head at a destination token D is represented as a weighted sum of result vectors from source tokens S, with the weights corresponding to the attention probabilities from D to S [22]. These vectors are then projected onto the unembedding vectors of their respective source tokens S, retaining only their negative components. Meanwhile, QK ablation involves mean-ablating the result vectors from each source token S, except for the top 5% of source tokens that are most likely to be predicted at the destination token D based on the logit lens. For instance, in the phrase "All's fair in love and war," if the destination token D is "and" and the token "love" is a highly predicted follower of D and appears as a source token S, the result vector from S is projected onto the unembedding vector for "love," and everything else is mean-ablated. This demonstrates how the attention head in question suppresses the prediction of "love." To evaluate the impact of the ablation, the token distribution output by the model for a given prompt (π) is compared with the distribution following an ablation (π_{Abl}) using KL divergence $D_{KL}(\pi||\pi_{Abl})$. By averaging these values over the OpenWebText dataset, D_{CSPA} for CSPA and D_{MA} for a mean ablation baseline are obtained. The proportion of the effect explained is then calculated as $1 - \frac{D_{CSPA}}{D_{MA}}$, with KL divergence chosen because a value of 0 indicates that the ablated and clean distributions are identical, implying that 100% of the head's effect is explained by the preserved components.